Benchmarking Windows 8 - by Mithun Mohandas

(Benchmark scores in bold are marginally better)

Test Rig

Processor......................Intel Core i7 - 3960X
Mobo...................ASRock X79 Extreme6/GB
RAM..............................Kingston HyperX 8GB (Dual Channel 1333 MHz)
SSD..........................Kingston KC100 120 GB
Graphics card.........................NVIDIA 650 Ti
Monitor....................................BenQ RL2450H
SMPS..........CoolerMaster SilentPro 850W
OS..Windows 8 Pro
....Windows 7 Ultimate

We started off by updating the motherboard BIOS (P2.10) to support Windows 8. Both operating systems were installed on similar Kingston HyperX SSDs with the same firmware (503). The latest drivers were obtained from the motherboard manufacturer's website and all care was taken to create the exact set of programs on both operating systems, and nothing unnecessary was installed.The benchmarks were chosen to bring out the differences in boot times, application performance, memory usage, file transfer, browser and gaming performance.

Gaming

Gamers needn't be worried when shifting to Windows 8. Games performed equally on both OSes, even 3DMark hit the same score during one of the runs. The only game that showed any difference at all was Metro2033, however, the average fps rules that out.

Synthetic

We used two all round testing suites, PC Mark 7 and GeekBench. According to GeekBench, Win 8 was found to be more hands on with memory management compared to Win 7 with nearly twice as much memory allocation instructions processed in the same amount of time. These results were reflected in the PC Mark results as well. Win 8 was also found to be consuming a bit more memory while idling.

File Copy

Writing assorted files onto external drives was slightly faster in Win 8.

Browser performance

Browser benchmarks were consistent on both operating systems as well.

Benchmark	Windows 7	Windows 8
Synthetic		
PC Mark 7	5139	5529
- Productivity	5078	4676
- Computation	4874	**11028**
- Storage	5269	5051
GeekBench	16200	16041
Idle memory use	1.6GB / 8GB	1.8GB / 8GB
3D Mark 11	X1448	X1448
- Graphics	1305	1305
- Physics	8186	8221
- Combined	1531	1530
Games		
Crysis 2	34.1	34.4
Metro 2033	14	13.33
SinperElite V2	9.2	8.7
File Copy Sequential		
- Read	1.33	1.31
- Write	1.31	1.3
File Copy Assorted		
- Read	1.4	1.53
- Write	2.31	**1.59**
Boot up		
Pre-Torture	31.22 sec	**22.15 sec**
Post-Torture	16 min 57sec	**11 min 57 sec**
Average	6 min 48 sec	**22.18 sec**
Browser		
Octane	16424	16566
Sunspider	140.4ms	139.7ms

Torture Test

After a fresh install, Windows 8 isn't that ahead of Windows 7 in terms of pure performance. So we had to simulate a few months of use in a short period of time. Fragger did the job of fragmenting the hard-drive wonderfully. In addition we installed 60K+ fonts, by the time all the fonts were installed the response times worsened. Idle memory usage the very first reboot after our "Torture test" showed a huge difference. Win 8 took 11 whole minutes to boot while Win 7 took an extra 6 minutes over Win 8, but after multiple reboots the Win 8 boot time had come down drastically to the earlier 22 second mark and Win 7 came down to about 7 minutes. During normal use after the torture test, the shutdown times for Win 8 were great but Win 7 had slowed down to a crawl.

soft's crown-jewel has differing avatars depending on the hardware platform of a particular device. Meaning, not only do you pick between Windows 8, Pro or Enterprise on your x86 PC, but if you want an ARM-based tablet or smartphone, you'll have to choose between either Windows RT or Windows Phone 8.

Obviously, there's expected to be a lot of confusion between the merits and demerits of the different Windows 8 flavours, and choosing the one best for you may not be easy. But you won't have to sweat over it too much, for that choice will be done for you by something else – unless you're contemplating between the different x86 versions of Windows 8. Devices will dictate the flavour of Windows you end up using and ultimately get stuck on, and there's no switching back unless you drop the device, flash its ROM or pick up a new one. Some of the devices featuring the different flavours of Windows 8 are quite promising. We've used and interacted with multi-touch display enabled notebooks, tablets and hybrids from Acer, ASUS, Dell, Fujitsu, HP, Lenovo and Samsung, and they all look promising. And don't forget Microsoft is also entering the hardware arena with its very own Surface RT and Pro tablets, sporting ARM and x86 flavours of Windows 8. We've played a bit with the Surface RT and it is by far the best Windows 8 tablet / slate that's out there at the moment. But expect, smartphone OEMs like Nokia, HTC, LG, Sony, Samsung and others to join the race very soon and offer competing Windows 8 tablets and hybrids. All through this year, you'll have OEMs racing to expand on their Windows 8 device portfolio and offerings.

What's in store

Predicting the future of Windows Vista was easy, even though it had an overhauled UI and some frustrating settings (remember UAC?). Even though Windows 8 may seem like Vista, bringing in a whole new UI, it isn't. It is a more mature operating system that is taking more than its fair share of risks. There is no doubt that Microsoft is late in the game in terms of coming out with a formidable opponent to take on the likes of iOS and Android. And where it has to appeal to the new, Windows 8 also has to stay relevant to everyone who's only ever used Windows all their life with a mouse and a large screen. The huge burden of legacy and tradition it carries just can't be sidelined. Yes, it isn't perfect for a desktop environment, there's way too much overlap between Modern and classic desktop UI modes. But if you look past the desktop, Windows 8 stands poised to take the world of tablets and hybrids by storm, if things go as per plan. Because in these emerging product segments – multitouch ultrabook, hybrids, slates and tablets – Windows 8 is a pleasure to use and may offer Microsoft more than a fighting chance to save it fortunes after all. The next 18 months hold the key to unlock its future, and we're hopeful.